

PERAK, Malaysia

WMO: 486200

Lat: 4.2211N

Lon: 100.7010E

Elev: 8

StdP: 101.23

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.4	22.8	20.8	15.5	26.9	21.5	16.2	26.6	5.1	29.5	4.4	29.8	0.6	70	0.307

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.7	33.9	26.8	33.2	26.7	33.0	26.7	28.0	32.2	27.7	31.7	27.4	31.4	2.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.7	30.8	26.5	22.0	30.1	26.2	21.6	29.8	90.0	31.9	88.2	32.2	87.0	31.7	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.9	4.3	3.7	DB	21.1	34.6	0.9	0.7	20.5	35.1	20.0	35.5	19.5	35.9	18.8	36.4	(4)
(5)			WB	20.3	28.8	0.9	0.9	19.7	29.5	19.1	30.0	18.6	30.6	17.9	31.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	27.4	27.7	28.2	28.4	28.5	28.3	28.0	27.9	27.7	27.6	27.4	27.3	(6)	
	DBStd	0.93	0.85	0.86	0.95	0.90	0.78	0.84	0.83	0.86	0.84	0.84	0.73	0.84	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6518	538	497	564	551	574	549	557	553	532	544	523	537	(10)	
	CDD18.3	3477	280	263	306	301	316	299	298	295	282	286	273	278	(11)	
	CDH23.3	35176	2634	2661	3211	3250	3504	3274	3123	3008	2794	2711	2486	2520	(12)	
	CDH26.7	13547	937	1054	1327	1334	1443	1366	1261	1176	1032	951	831	835	(13)	
(14) Wind	WSAvg	1.2	1.1	1.2	1.2	1.3	1.2	1.1	1.2	1.2	1.2	1.2	1.1	1.0	(14)	
(15) Precipitation	PrecAvg	1726	131	135	108	150	119	81	95	103	170	217	222	191	(15)	
	PrecMax	2102	256	297	361	294	251	189	253	257	311	564	376	354	(16)	
	PrecMin	1379	8	31	23	9	35	12	7	28	31	89	104	85	(17)	
	PrecStd	188	67	80	72	66	64	42	51	53	65	84	78	74	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	33.9	34.2	34.2	34.1	34.1	33.9	33.9	33.4	33.2	32.7	32.9	(19)	
		MCWB	26.3	26.4	27.0	27.3	27.1	26.8	26.5	26.2	26.2	26.4	26.9	26.7	(20)	
	2%	DB	32.2	33.0	33.2	33.2	33.2	33.2	33.0	33.0	32.7	32.2	32.0	32.0	(21)	
		MCWB	26.1	26.1	26.7	27.1	27.1	26.8	26.5	26.3	26.3	26.3	26.6	26.3	(22)	
	5%	DB	31.6	32.2	32.8	32.8	32.8	32.8	32.2	32.2	32.0	31.8	31.2	31.2	(23)	
		MCWB	26.0	26.0	26.6	26.9	27.1	26.7	26.4	26.2	26.2	26.3	26.5	26.2	(24)	
	10%	DB	30.9	31.6	32.0	32.0	32.1	32.1	31.8	31.6	31.2	30.9	30.8	30.7	(25)	
		MCWB	25.9	25.9	26.4	26.7	26.9	26.6	26.3	26.1	26.1	26.2	26.3	26.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.6	27.6	28.1	28.2	28.5	28.2	27.8	27.6	27.6	27.5	27.8	27.6	(27)	
		MCDB	31.4	32.1	32.8	32.5	32.7	32.5	32.1	32.0	31.7	31.3	31.5	31.6	(28)	
	2%	WB	26.9	27.0	27.5	27.7	28.0	27.6	27.3	27.1	27.0	27.0	27.2	27.0	(29)	
		MCDB	30.7	31.3	31.9	31.8	31.8	31.7	31.4	31.2	30.9	30.8	30.7	30.7	(30)	
	5%	WB	26.5	26.6	27.1	27.3	27.6	27.2	26.8	26.7	26.6	26.6	26.7	26.6	(31)	
		MCDB	30.1	30.6	31.3	31.3	31.3	31.2	30.8	30.6	30.4	30.3	30.1	30.0	(32)	
	10%	WB	26.1	26.2	26.7	27.0	27.2	26.8	26.5	26.4	26.3	26.2	26.4	26.2	(33)	
		MCDB	29.5	30.1	30.6	30.8	30.9	30.7	30.4	30.2	29.9	29.7	29.6	29.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	8.3	8.2	7.8	7.7	8.0	8.0	7.9	7.6	7.3	7.1	7.2	(35)	
		MCDBR	8.5	9.0	8.9	8.5	8.5	8.7	8.7	8.7	8.5	8.2	7.8	8.0	(36)	
	5% WB	MCWBR	3.5	3.5	3.3	3.2	3.3	3.4	3.4	3.3	3.2	3.2	3.2	3.2	(37)	
		MCWBR	7.8	8.3	8.3	8.0	7.9	8.1	8.2	8.1	7.8	7.6	7.4	7.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.470	0.514	0.515	0.501	0.479	0.507	0.501	0.509	0.513	0.534	0.475	0.461	(40)	
		taud	2.285	2.135	2.161	2.205	2.279	2.205	2.213	2.178	2.167	2.104	2.285	2.328	(41)	
	Ebn at Noon		855	825	823	818	816	781	791	801	812	798	844	856	(42)	
		Edh at Noon	137	162	158	149	134	142	142	151	155	165	136	130	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.81	5.47	5.59	5.54	5.17	5.12	5.02	5.10	4.90	4.77	4.60	4.45	(44)	
	RadStd		0.37	0.35	0.36	0.18	0.21	0.22	0.21	0.21	0.34	0.40	0.22	0.32	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		+0.44	+0.31	+0.26	+0.56	+0.47	+0.50	N/A	N/A	+162	+162	(46)	
(47) Regional (4 neighbors)		+0.39	+0.22	N/A	+0.52	+0.44	+0.37	N/A	N/A	+140	+140	(47)	

Nomenclature: See separate page